

ENVIRONMENTAL SEGMENTS

Broadly speaking, the environment consists of the following four segments :

Atmosphere

Hydrosphere

Lithosphere

Biosphere

These segments are discussed below:

(i) **Atmosphere:** Atmosphere extends up to about 500 kms above the earth's surface. The atmosphere comprises of a mixture of gases like nitrogen and oxygen (major components), and argon, carbon dioxide (minor components) etc.

Oxygen is essential for life on earth, Carbon-dioxide is essential for photosynthesis. Nitrogen is used by nitrogen fixing bacteria and ammonia-manufacturing plants to yield chemically bound nitrogen essential for life. The temperature, pressure and density of the atmosphere vary with altitude. Depending upon the altitude the temperature of the atmosphere varies from - 100° C to + 1200° C. The atmospheric pressure is 3×10^7 atmosphere at 100 Km above the sea level and it rises to 1 atmosphere at sea level.

It also helps to maintain the heat balance of earth through absorption of radiations emitted by Sun and reemitted from the earth. It also protects the earth from dangerous cosmic radiations from outer space and helps in sustaining life on earth through a phenomenon known as "Green-house effect". Atmosphere is also a vital carrier of water from oceans to land (as part of hydrologic cycle).

(ii) Hydrosphere: Hydrosphere constitutes About 80% of the surface of the earth which is covered by water and because of this earth is also known as blue planet. This includes all types of water resources-surface as well as ground like lakes, streams, rivers, seas oceans etc. and ground water, water locked in rock-crevices and minerals lying deep below the earth's crust. It is important to note that not even one percent of the total world's water resources is available for exploitation by man for industrial, agricultural and domestic purposes. This is due to the fact that water from 'oceans and seas is too saline to drink and for the direct agricultural and industrial uses. Without water life on earth would have been impossible.

(iii) Lithosphere: it is the rigid, rocky outer layer of the earth consisting of the crust and the solid outermost layer of the upper mantle. It extends to a depth of about 100 km.

The most important part of lithosphere is soil. This essentially consists of complex mixture of inorganic and organic matter and water. The mixture of silicates (of sodium, potassium, calcium, aluminium and iron); oxides (of iron, manganese and titanium) and carbonates (of calcium and magnesium) constitute inorganic matter.

The organic matter is present up to about 5 percent and it mainly determines the productivity of the soil. Polysaccharides nucleotides, organo-phosphorous and organo-sulphur compounds, sugar and humus constitute organic matter content of the soil. In fact, lithosphere is the outer surface of the earth and it consists of earth's crusts and soil.

(iv) Biosphere: Biosphere covers the entire realm of living organisms and their interactions with the other segments of the environment. In fact, biosphere is thin life supporting stratum of the earth surface, extending from a few miles into the atmosphere to the deep sea layers of the oceans.

It is the region of the earth where life exists and is a self regulating system. Broadly, biosphere is divided into two parts abiotic and biotic.

COMPOSITION OF ATMOSPHERE

The atmosphere is a blanket of air around the earth. Air is a mixture of gases. In homosphere (Lower 80 km zone of atmosphere) the proportion of various gases is fairly constant, except for varying amounts of water vapour (which is restricted to troposphere). In heterosphere, the gases are present in insufficient quantity failing to form a mixture similar to air. The maximum concentration of atmospheric gases lie in the lower atmosphere due to the forces of gravity.

The concentrations of the components of the atmosphere are given in table 1.

Table 1: Composition of Dry Unpolluted Air by Volume

Gas	Concentration
Nitrogen	78.084%
Oxygen	20.946%
Argon	0.934%
Carbon	340 ppm
Neon	18.18 ppm
Helium	5.24 ppm
Methane	1.3-1.6 ppm
Krypton	1.14 ppm
Hydrogen	0.5 ppm
Nitrous oxide -	0.25-0.35 ppm
Xenon	0.087 ppm

LAYERS OF ATMOSPHERE

The atmosphere can be broadly divided into four major regions viz. Troposphere, stratosphere, Mesosphere and Thermosphere. The important characteristics of these regions are summarized in table 2. and Fig. 1.

Name of the region	Height above earth's surface	Temperature range (°C)	Major chemical species present
Troposphere	0-11	15 to -56	O ₂ , N ₂ , CO ₂
Stratosphere	11-50	-56 to -2	O ₃
Mesosphere	50-85	-2 to -92	O ₂ ⁺ , NO ⁺
Thermosphere	85-500	-92 to 1200	O ₂ ⁺ , O ⁺ , NO ⁺

Scanned figure is given in the last.

(a) Troposphere:

Essentially all the atmosphere's water vapour is in the troposphere, which, therefore, encloses essentially all the storms and precipitation of the earth. In the absence of any significant air pollution, the composition of air in troposphere remains uniform due to convection currents arising from differential heating and cooling rates between the equator and the poles.

In the troposphere, the temperature of the air decreases with increasing altitude up to 'tropopause' where the temperature starts increasing with increasing altitude in the stratosphere. The change of temperature with height is called the lapse rate. Positive lapse rate is the decrease of temperature with increasing altitude (in the troposphere). Temperature

inversion occurs at the tropopause where the transition from positive lapse rate to negative lapse rate takes place.

(B) Stratosphere:

Stratosphere is the region above the tropopause. In this region, the temperature-altitude curve exhibits a negative lapse rate. The temperature in this region continues to increase with height due to the absorption of ultraviolet radiation from sun by ozone. At the top of the stratosphere there is a narrow zone known as stratopause where the temperature begins to fall again with increasing height as the mesosphere is entered:

(C) Mesosphere:

Mesosphere is the region above the stratopause. In this region, the temperature-altitude curve exhibits a positive lapse rate. The temperature in this region continues to decrease with height due to the absence of species like ozone which can absorb u. v. radiations from the sun. The temperature attains a minimum at the mesopause, a thin zone of coldest temperature where the temperature stabilizes and soon start to rise again as the thermosphere is ascended.

(d) Thermosphere:

Thermosphere is the region above the mesopause. In this region, the temperature-altitude curve exhibits a negative lapse rate. The temperature in this region increases very rapidly with increasing altitude. This region is characterised by low pressure, low densities and high temperatures (about 1200°C).

Oxygen and nitric oxide absorb the solar radiations in the far UV region and undergo ionisation.

It is important to note that apart from changes of temperature with height, changes in chemical composition or activity also occurs.

For instance, in chemosphere (a region which includes whole of stratosphere, mesosphere and lower part of the thermosphere) major chemical changes occur. Mainly photochemical reactions take place in this region. The most common is the photodissociation of oxygen resulting in the formation of ozone

Green house effect/ global warming

The earth is heated by sunlight. Earth radiates some of this heat back into space. However, some gases like carbon dioxide, water vapours, methane and chloro fluoro carbons in the lower atmosphere (acting like glass in a greenhouse) allow solar radiation to come in but these gases strongly absorb the infrared radiations which the earth sends back as heat. A part of the heat so trapped in these gases is re-emitted to the earth surface. The effect of radiation of thermal energy back to the earth by the atmosphere is called Green House effect. In fact, this green house keep our planet house effect keeps our planet and thus helps in sustaining life on earth because in the absence of this effect, all the heat radiated by the earth would be lost to the space and the earth would become very cold .

In recent time the condensation of green house gases increased at a very fast rate due to industrialisation ,deforestation, and increased burning of fossil fuels. The increased concentration of GHG , especially carbon dioxide enhance green house effect leading to increase surface temperature. The increase earth temperature is responsible for the global warming.

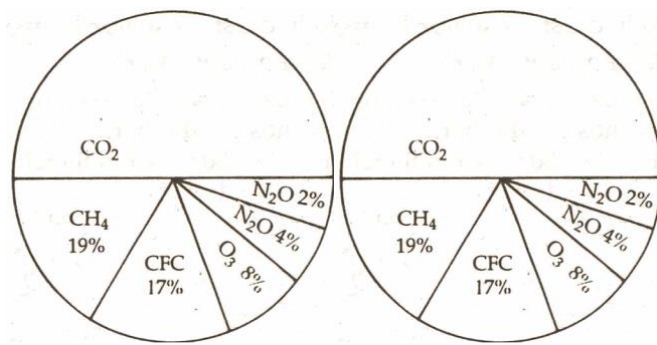


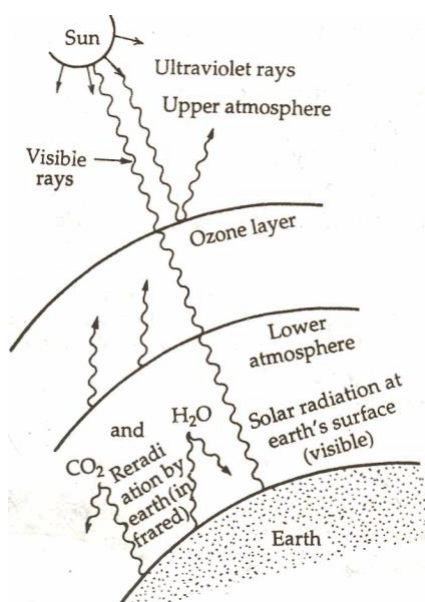
Fig. Relative contributions of active gases to temperature rise.

consequences of Green house effect:

- The surface water would evaporate faster than normal on slight increase in the surface temperature, creating drought.
- Global warming or Green house effect can adversely affect world's food production. This is due to the alterations in the pattern of weather leading to crop damage and/ or crop failure.
- As a result of green house effect there would be a rise in sea level. An increase in the average surface temperature of the earth could melt most of the polar ice caps of Green land and Antarctica resulting in the flooding of many of the low level coastal areas.
- Due to the warming of surface layer, biological productivity of the ocean would also be decreased which in turn reduce transport of nutrient from the deeper layer to surface by vertical circulation.

DEPLETION OF OZONE LAYER

A layer of ozone is constantly formed in the stratosphere from oxygen gas by the absorption of ultraviolet light. The ozone layer acts as a protective shield for the existence of life on earth. It absorbs most of the harmful ultra-violet radiations coming from the sun in the region (220 - 330 nm) and thereby protects the life from several radiation damages.



Importance of O₃ layer

It is found that during past decades due to increased air pollution there is an overall thinning of ozone layer everywhere especially above Antarctica and arctic, this thinning of ozone layer is called ozone depletion.

Chloroflouro carbon compounds are found to be main reason of ozone depletion.

CFC are chemically stable ,odourless, noninflammable ,non-toxic and noncorrosive compounds seems to be dream chemicals .Being cost effective they become popular as coolant in air conditioners, refridgerators,propellent in the aerosol spray cans, cleaners for electronic parts such as computer chip, sterilants for hospital instruments, fumigant for granaries and ship cargo holds etc.

How these CFCs are responsible for the depletion of ozone layer can be understood as follows:

CFCs remain in the troposphere as they are insoluble in water and chemically inactive Over 11-12 years they rise in to stratosphere mostly through convection ,random drift, and turbulent mixing of air in the

troposphere. Once they reach the stratosphere CFC molecules break down under the influence of high energy UV radiation. This releases highly reactive chlorine atoms, which speeds up the breakdown of reactive ozone into O_2 and O in a cyclic chain of chemical reactions. This causes ozone in various parts of the stratosphere to be destroyed faster than it is formed. Finally each molecule can last in the stratosphere for 65-385 years depending on its types. During that time each chlorine atom released from these molecules can convert up to 100,000 molecules of O_3 to O_2 .

Ozone can also be destroyed by its reaction with nitric oxide, hydroxyl radical etc. present in atmosphere due to

- (i) chemical and photochemical reactions in the atmosphere, nuclear explosions, supersonic jets, etc. generate Nitric Oxide (NO) ;
- (ii) chlorine is released in atmosphere due to volcanic activity.
- (iii) bio-mass burning leads to $OH^\cdot$ generation

In the absence of ozone layer, these ultraviolet radiations would cause

- diseases like skin cancer in men and animals, swelling of skins, burning sensation,
- skin aging, leukaemia, breast cancer, cataracts of eyes, haemorrhage, lungs injury,
- premature aging, visual impairment, dizziness, inhibition and alteration of DNA's
- replication and formation of DNA adduct, and may even lead to Death.
- DNA mutation,
- damage the plants, and
- faster deterioration of paints, fabrics, plastics etc.

AIR POLLUTION

Air pollution may be defined as the presence of one or more pollutants (or combination thereof) in the outdoor atmosphere in such quantities and of such durations, which might be injurious to human, plant or animal life or property. These pollutants may unreasonably interfere with the comfortable life or the conduct of business.

AIR POLLUTANTS

Air is never found clean in nature, due to natural and manmade pollution. Gases such as CO, SO₂ and H₂S are continually released into the atmosphere through natural activities, e.g. volcanic activity, vegetation decay and forest fires. Besides, tiny particles of solids or liquids are distributed throughout the air by winds, volcanic explosions and other similar natural disturbances. In addition to these 'natural pollutants', there are manmade pollutants-gases, mists and particulates, aerosols resulting from the chemical and biological processes used by man. The latter are present in relatively high concentrations compared to the background values in air. These pollutants hardly exist beyond 2000 feet above ground level. Air pollutants are present in the atmosphere in concentrations that disturb the dynamic equilibrium in the atmosphere and thereby affect man and his environment.

Causes of air pollution;

- Population explosion
- Industrialisation
- Urbanisation
- Automobiles

- Natural phenomena like volcanic eruption, radioactivity, forest fires etc.

Depending of their origin pollutants are divided into two categories

- Primary pollutants and
- Secondary pollutants

Primary Pollutants; pollutants which are directly emitted into the atmosphere There are five primary pollutants which together contribute more than 90% of global air pollution. These are:

- Carbon monoxide, CO
- Nitrogen oxides, NO_x
- Hydrocarbons. HC
- Sulphur oxides, SO₂
- Particulates, part

Secondary pollutants; pollutants which are derived from the primary pollutants due to chemical and photo chemical reactions in the atmosphere, eg. Ozone, Peroxy-acyle nitrate, photochemical smog etc

According to chemical composition;

Organic pollutants, e.g.

- Hydrocarbons
- aldehyde
- ketones
- alcohols etc.

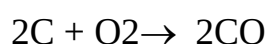
Inorganic pollutants;

- carbon compounds
- Nitrogen compounds
- Sulphur compounds
- Halogen compounds
- Oxidising compounds

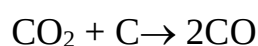
Carbon Monoxide

It is a colourless, odourless and tasteless gas above -192°C (-314°P). It is 96.5% as heavy as air and is not soluble in water. The basic chemical reactions yielding CO are:

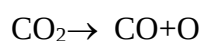
- Incomplete combustion of fuel or carbon-containing compounds:



- Reaction between CO_2 and carbon containing materials at elevated temperatures in industrial processes, e.g. in blast furnaces



- . Dissociation of CO_2 at high temperatures



Sources and Sinks of CO Pollution

Natural processes, e.g. volcanic action, natural gas emission, electrical discharge during storms, seed germination, marsh gas production, etc. contributes in a small measure to CO in the atmosphere. The significant contribution is from human activities.

The annual emission on a global scale is 350 million tonnes (human source 275 and natural 75 million tonnes), of which the USA alone releases more than 100 million tonnes of CO into the atmosphere. (a) Transportation contributes about 64% of CO-motor vehicles, 59.2% aircrafts 2.4%; and railroads 0.1 %. (b) Next in magnitude is miscellaneous sources, 16.9%-the main components are forest fires, 7.2% and agricultural burning, 8.3%. Agricultural burning implies controlled burning of forest debris, crop residues, brush, weeds and other vegetation. (c) Industrial processes, mainly iron and steel industries and petroleum and paper industries, constitute the third largest contributor of CO (9.6%) to the air. The atmospheric background of CO is 0.1 ppm.

Sinks-The annual input of CO into the atmosphere by human activities is expected to double its concentration in the ambient (surrounding) atmosphere every five years. But the actual increase in ambient global CO concentration is much less. The major CO sink is some soil micro-organisms. A potting soil sample weighing 2.8 Kg completely removed, in three hours, 120 ppm CO from ambient air.

The same soil sample, when sterilised, failed to remove CO from air.

Of 200 micro-organisms isolated from the soil and cultured, 16 fungi were found to be active in the removal of CO.

The soil sinks can take care of atmospheric CO, but it still exists in significant concentrations in the atmosphere. This is because neither CO nor the soil sink is distributed uniformly. As a matter of fact, the largest CO producing areas often have the least amount of soil sink available.

Since the automobile remains the largest single source of CO pollutant (59.2%), highly populated urban areas have the highest ambient CO concentrations. Figure 6.1 shows the typical CO concentration profile in a busy downtown street of Manhattan, USA.

CO concentrations, averages over 8-hour periods, are given in Table 6.2 for locations with increasing distance from traffic. An 8-hour period is used as the body absorbs and desorbs CO slowly. At low levels of CO, 4 to 12 hours are required so that an equilibrium can be established between the concentration of COHb in the body and the average CO concentration of the inhaled air. The first two locations in Table 6.2 are applicable to taxi, and bus drivers and traffic policemen, while the last two locations apply to the average citizen.

Control of CO Pollution

The transportation sources are responsible for 74% of all CO emissions and gasoline-fed internal combustions (Table 6.1) are primarily accountable for it. Hence, the considerable efforts for control of CO pollution are mainly directed automobiles. Use of catalytic converters in two stages helps in eliminating pollutants from exhaust gases before they are discharged into the atmosphere.

In the first converter $(\text{NO})_x$ is reduced to N_2 (+ NH_3) in the presence of finely divided Pt, as catalyst, and the reducing gases, CO and hydrocarbons. The production of NH_3 is kept at a minimum under carefully controlled conditions.

In the second converter, air is introduced to provide an oxidising atmosphere for complete oxidation of HC and CO into H_2O and CO_2 in the presence of finely divided Pt catalyst. One problem of the Pt catalyst is that it is liable to be poisoned by heavy metals (e.g. Pb) present in gasoline itself. Hence Pb-free gasoline must be used for proper function of catalytic converters.

CO level (ppm)	Effects on humans
10	Impairment of judgement and visual perception
100	Headache, dizziness, weariness
250	Loss of consciousness
750-1000	Death

The soil micro-organisms act as a major sink for carbon monoxide from ambient atmosphere. But still it exists in significant concentrations in the

atmosphere, because the largest CO-producing areas have the least amount of soil sink available around them.

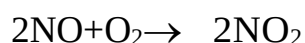
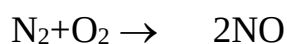
The following efforts have been directed towards controlling carbon monoxide emission from transportation section:

- Fuel modification
- Development of substitute fuels,
- Modification of engine design
- Treatment of exhaust gases.

Oxides of Nitrogen (NO_x)

NO and NO₂ are mostly responsible for air pollution and they are jointly represented as NO_x is formed by photolytic reactions in the atmosphere. Oxides of nitrogen may also be formed by natural or artificial fixation of nitrogen from the atmosphere or from nitrogen compounds present in organic matter. NO_x can also be produced by the combustion of coal, oil, natural gas and other organic matter. hence, NO_x is introduced into the atmosphere from incinerators, automobile exhausts, coal based power plants etc. Some of the involved reactions are:

1210 - 1765° C

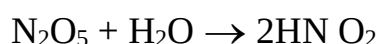
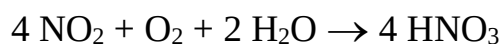
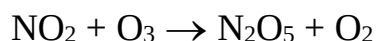
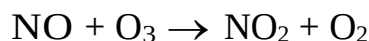


The most important primary photochemical reaction is the dissociation of NO₂,

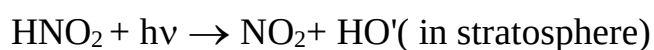
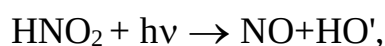
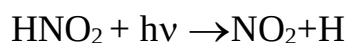


The NO so formed may be oxidized by O₃ (or more slowly by O₂), thus leading to cyclic chain reaction. When the NO₂ is completely converted

into HNO_3 by hydration and catalytic oxidation in presence of aerosols, fog or photochemical smog, the rain may be broken.



HNO_2 and HNO_3 may also undergo photochemical dissociation as follows:



The nitric acid (HNO_3) so formed is removed as acid rain because it gets precipitated as nitrates during rainfall. It is also important to note that photochemical and chemical reactions involving NO_2 and hydrocarbons induced by sunlight can lead to the formation of photochemical smog.

The following efforts have been directed in controlling NO_x emissions from various sources:

- (a) Use of two stage catalytic converters in automobiles,
- (b) Use of a two-stage combustion process in power plants

Sulphur dioxide

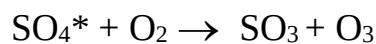
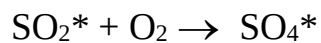
(SO_x) A mixture of SO_2 and SO_3 is commonly represented as SO_x

The sources of SO_x pollution are summarized below:

- Volcanic activity,
- Combustion of any sulphur-bearing materials, say in coal-fired power stations

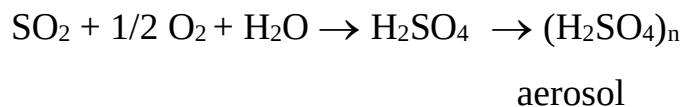
- and refineries,
- Transportation,
- Smelting of sulphide ores in metallurgical operations,

Atmospheric sulphur dioxide absorbs solar radiation in the range of 300-400 nm and produces electronically excited states of SO_2 , { $\text{SO}_2 \rightarrow \text{SO}_2^*$. This undergoes oxidation to SO_3



SO_3 in presence of water vapour gets converted to H_2SO_4 .

The overall reaction may be represented as



The above reaction is catalysed by olefinic hydrocarbons and oxides of nitrogen. The droplets of H_2SO_4 aerosol can give rise to acid-rain. H_2SO_4 and $(\text{H}_2\text{SO}_4)_n$ aerosols present in the urban air can easily reach the pulmonary region of lungs, causing serious respiratory problems (particularly in older peoples).

The following efforts have been directed in controlling

SO_x emissions from the anthropogenic activities:

- The use of Low-sulphur fuel, , '
- Discouraging power generation by fossil-fuel based thermal power plants

Hydrocarbons

There are mainly three sources which emit hydrocarbons in the atmosphere. They are

- solvent evaporation
- automobile exhausts and
- burning of wood, coal, oil and refuse

Methane is mainly emitted into the atmosphere by natural activities such as anaerobic decomposition of organic matter in Water, soil and sediments by micro organisms. Ethane, n-butane, n-pentane etc. 3.rle other hydrocarbons which have been detected in atmosphere in areas of heavy vehicular traffic.

These hydrocarbons gets oxidised in the atmosphere by a series of chemical and photochemical reactions. This results in formation of CO₂, solid organic particulates and water soluble acids and aldehyde.

The hydrocarbons also lead to the formation of photochemical smog.

Particulates

Volcanic eruptions, wind and dust storms, salt spray etc. are natural sources of particulate constituents of the atmosphere.

Burning of coal, oil, wood, gaseous fuels; Industrial processes; Smelting and mining operations; fly-ash emissions from power plants , burning of coal refuse and agricultural refuse are man-made activities which release considerable tonnage of particulate matter in atmosphere.

Particulate matter has large surface areas hence they provide excellent sites for absorption of various organic and inorganic species .They also scatter light and reduce visibility.The typical particulate air pollutants their sources and effects are summarized below in;

Particulate	Sources	Effects
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Dust	Asbestos and cement factories	Causes respiratory and allergic diseases
Smoke	In complete oxidation of fuels in automobiles	Causes skin diseases
Lead dust	lead batteries and automobile exhausts	Causes lead poisoning

Particulate matter can be removed from industrial gaseous effluents with the help of following devices; cyclone collectors, electrostatic precipitators, wet scrubbers and settling chambers.

Indoor pollution

Indoor pollution is defined as "the presence of physical, chemical or biological contaminants in the air of confined environments, which are not naturally present in high quantities in the external air of the ecological systems."(Italian ministry for the Environment, 1991)

In the last thirty years much attention has been paid to reducing the outdoor pollution, but only recently has the international scientific community worried about reducing the contamination of the air of closed environments. If we consider the amount of time a person spends in a closed environment (90%) we will understand that the issue of indoor pollution is of primary importance.

The atmospheric composition inside an edifice is fundamentally the same we find outside, but the amounts and types of contaminants differ. To the pollutants present outside one must add all the polluting agents generated within the edifices.

In the developing countries It is the rural area that face the greatest threat from indoor pollution, where the people continue to rely on traditional fuels such as firewood, charcoal, and cow dung for cooking and heating. Concentration of indoor pollutants in households that burns tradition fuels are alarming'

Burning such fuel produce large amount of smoke and other air pollutants in the confined space of home resulting in the high exposure. Women and children are the most vulnerable group as they spend more time indoor and are exposed to smoke.

In urban area , exposure to indoor pollution has increased due to variety of reasons including construction of more tightly sealed buildings, reduced ventilation, use of synthetic materials for building and furniture and use of chemical products, pesticides ,and household care products .

The main sources of indoor pollutants are:

- Construction materials
- Heating, air-conditioning devices, and cooking apparatuses etc.
- Furniture
- Coatings (wall paint, varnish, floors etc.)
- Maintenance and cleaning products (detergents, pesticides etc.)
- Use of space and activities done within it

The study of the effects of air pollution on the human health is complex for the symptoms are not specific and many pollutants can be responsible for the same disease. Furthermore there are also the effects caused by stress and climatic discomfort. The fact that different people react differently to the same conditions must be considered as well.

Following are the some of the main compounds for indoor pollutants

- Volatile organic compounds
- Pesticides
- Biological pollutants
- Formaldehyde
- Asbestos
- Radon

The main observed effects are:

- Respiratory
- Dermal and mucous irritation
- Effects on the nervous system
- Cardiovascular
- Effects on the gastrointestinal apparatus
- Effects on the reproductive system
- Infections and intoxication

Earth's Radiation Budget

Most input of the Earth energy is received from the Sun. The solar energy is short-wave radiation. Although the Earth also receives electromagnetic energy from the other bodies in space, it's negligible, compared with solar energy. The incident solar energy (short-wave) may be reflected and absorbed by the Earth's surface or the atmosphere. And Earth's surface and

atmosphere also emit the radiation (longwave). The Earth Radiation Budget is the balance between incoming energy from the sun and the outgoing long wave (thermal) and reflected short-wave energy

The Earth's Radiation Budget is a concept used for understanding How much energy the Earth gets from the Sun and How much energy the Earth-system radiates back to outer space as invisible light.

Absorbed sunlight raises the Earth's temperature. Emitted radiation or heat lowers the temperature. When absorbed sunlight and emitted heat balance each other, the Earth's temperature doesn't change - the radiation budget is in balance

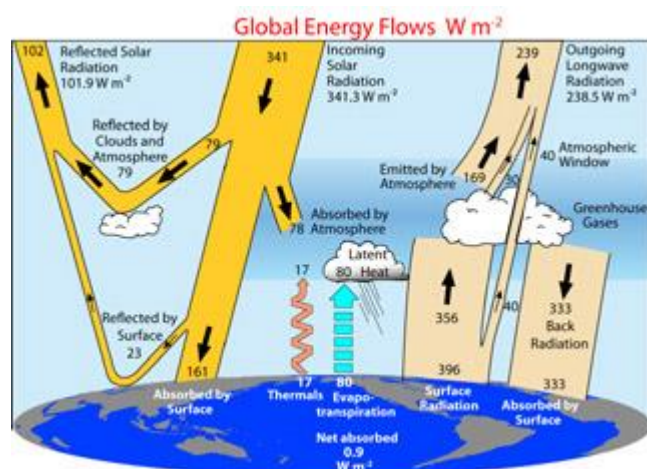
Light from the Sun shines on Earth. Some of that light reflects off clouds back into space. Some of the light makes it to the ground and warms our planet. The warm ground and oceans give off infrared (IR) radiation, which we feel as heat. That IR "light" moves back up through the atmosphere. Most of it is trapped by greenhouse gases, which keep Earth warm. After a while, the IR radiation leaks back out into space. Scientists call this cycle of incoming and outgoing energy "Earth's radiation budget".

The picture on this page shows a lot of details about Earth's radiation budget. This kind of radiation isn't the kind from atom bombs or nuclear power plants. Instead, it is electromagnetic radiation. It is mostly visible light and infrared radiation. Scientists measure it in units of watts per square meter (W/m^2 or $\text{W}\cdot\text{m}^{-2}$). Imagine laying out a one meter by one-meter square on the ground. Now measure how much solar energy falls on that square each second. That's what we're talking about here. The average energy from sunlight coming to the top of Earth's atmosphere is around 341.3 W/m^2 .

Less than half of the incoming sunlight heats the ground. The rest is reflected away by bright white clouds or ice or gets absorbed by the atmosphere. The warm ground gives off IR radiation. The right side of the picture shows that. Our atmosphere has greenhouse gases in it. They let normal light pass through, but trap IR "light". It is sort of like having a blanket covering Earth. Our planet would be about 30°C (54°F) colder if there wasn't a greenhouse effect! After a while, the IR "light" slowly leaks back out into space.

For the most part, the energy coming to Earth as sunlight equals the energy leaving as IR "light". If it didn't, Earth would heat up or cool down. These days the balance is a little off. Humans have added lots of greenhouse gases to the atmosphere. Earth's climate is slowly warming up. That's why

scientists are really interested in Earth's radiation budget. It helps them understand global warming



Smog-photochemical and sulphurous

The name smog is derived from a combination of smog and fog. This is of two types.

1. Classical Smog (sulphurous smog) or London smog
2. Photochemical smog or Los Angeles smog

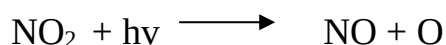
In the presence of smog, the atmospheric visibility is partially obscured by a haze consisting of solid particulates and/or liquid aerosols.

Photochemical Smog

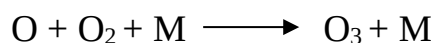
Mechanism

Formation of hydroxyl radical

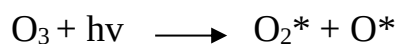
The NO_2 absorbs visible and UV radiations from sunlight in the atmosphere. So the photolysis takes place and ground state oxygen is formed.



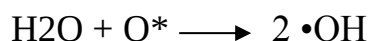
This atomic oxygen reacts rapidly with O_2 in the presence of a third body M (O_2 or N_2) and produce ozone.



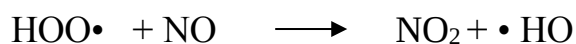
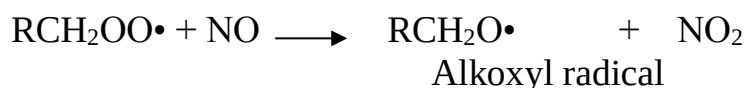
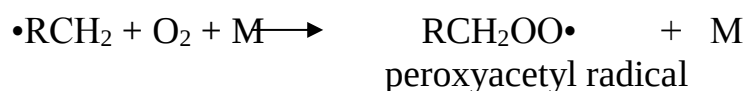
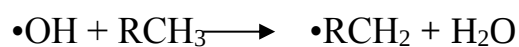
This ozone undergoes photolysis by UV radiations from sunlight to form oxygen molecule and oxygen atom, both in the excited state.



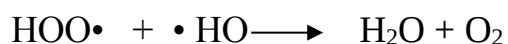
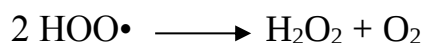
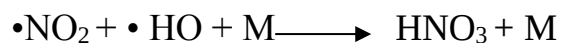
It reacts with water vapour to form $\cdot\text{OH}$ radicals.



Oxidation of hydrocarbons

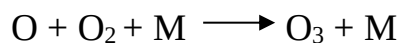
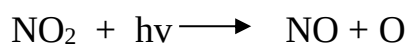


In order to reduce the concentration of hydroxyl radicals from the atmosphere, several termination reactions act to remove $\cdot\text{HO}$ radical, its precursor NO_2 and the $\text{HOO}\cdot$ radical.

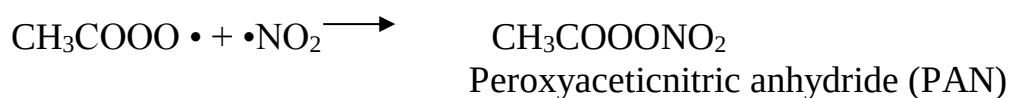
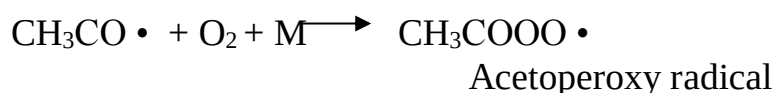
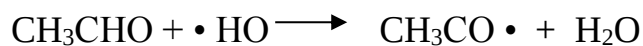


The nitric acid and hydrogen peroxide are water-soluble and are removed from the atmosphere by precipitation.

Some secondary reactions are also occurred simultaneously with the oxidation of hydrocarbons. Like



Other reactions as CH_3CHO an example

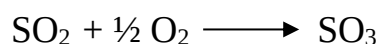
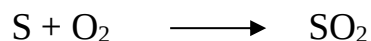


Sulphurous Smog

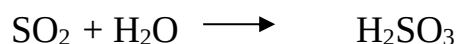
Sulphurous smog consists of carbon-based soot, other solid particulates and high levels of SO_2 . The most important source of this smog is combustion of coal. The other sources are automobile exhausts, industries, burning of wood and agricultural waste. The most severe air pollution due to sulphurous smog was first observed in London, in 1952. So this is also called London smog.

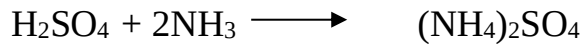
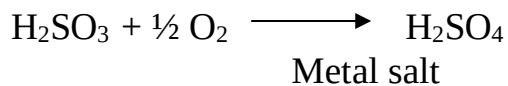
Mechanism

It is generally worst in early hours of morning and appears to worsen shortly after sunrise, perhaps because of photochemically induced oxidation of SO_2 into sulphur trioxide followed by combination with moisture to give aerosol.

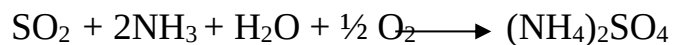


The second reaction takes place in the presence of sunlight or particulate matter. Sulphur dioxide combines with moisture from fog to form sulphurous acid (H_2SO_3), the oxidation of which is very rapid and more rapid in the presence of metal salts.





or SO_2 combines with NH_3 in moist air.



Harmful effects

Sulphurous smog produces more dangerous effects than the sum total of SO_2 and particulates. Sulphate haze is considered as a greenhouse gas. Now, this is the reason of climate cooling. It creates acid rain and depletes ozone layer. This smog is characterized by bronchitis irritation.

Preventive measures

The control of primary pollutants, such as NO_x and hydrocarbons will ultimately control the secondary pollutants. The techniques, like incineration, absorption, adsorption and condensation are used to control hydrocarbon emissions from stationary sources.

1. Incineration method

For hydrocarbon removal the flame after burner is used, in which flame is used to complete the oxidation of hydrocarbon into CO_2 and H_2O . The other device is used for this, is the catalytic after burner in which the catalyst oxidizes the hydrocarbon at a lower temperature.

2. Adsorption method

In this method, exhaust gases are passed through a bed of granulated adsorber consisting of activated carbon. Hydrocarbon vapours are adsorbed on the surface of activated carbon. The hydrocarbons are then condensed to liquids and can be used for further purpose.

3. Absorption method

Liquid containing dissolved hydrocarbon is brought in contact of exhaust gases. The contact between absorbing liquid and exhaust gases is carried in tall scrubbing towers.

4. Condensation method

Smog forming pollutants are condensed to liquids, which are collected and reused.

The control of hydrocarbon emissions coming out from automobiles is carried out by the use of oxidizing catalyst, which convert CO and hydrocarbons to CO₂ and H₂O while the reducing catalyst convert NO_x to N₂.

Acid Rain

Normal rainwater is slightly acidic because of that, CO₂ present in the atmosphere gets dissolved in it and form carbonic acid, H₂CO₃. But in the atmosphere, the SO₂ and NO₂ gases are also present as air pollutants. These gases forms the sulphuric acid and nitric acid and further, lowered the pH of the rain. When the pH of rain is 5 or below, is known as acid rain.

Mechanism

Acid rain can either be wet or dry. Acid rain, snow, dew, fog, frost and mist are the wet form of deposition, while dust particles containing sulphates and nitrates are called dry deposition. (Mechanism as in sulphurous smog and in photochemical smog)

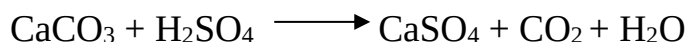
Factors responsible

Factors responsible for acid rain are mainly human activities – cars, homes, factories, power stations, transport and subsequent deposition of oxides of sulphur, nitrogen and their oxidative products. Other factors are volcanoes, combustion of fossil fuels, smelters, swamps and planktons in the ocean.

Effects

Acid rain affects on materials, terrestrial ecosystems and aquatic ecosystems etc. Acids degrade building materials, especially limestone, marble, various commonly used metals such as galvanized steel and certain

paints. The increased rate of weathering and erosion of building surfaces and monuments are also the adverse impacts of acid rain.



Air Quality Standards

The Clean Air Act requires the EPA to establish both air quality standards (NAAQS) and emission standards (NSPS). National Ambient Air Quality Standards (NAAQS) are acceptable concentrations of pollution in the atmosphere, while emission standards (NSPS) are allowable rates at which pollutants can be released from a source.

EPA has established the National Ambient Air Quality Standards (NAAQS) at two levels: primary and secondary.

Primary standards are set at levels that will protect public health and include an “adequate margin of safety,” regardless of whether the standards are economically or technologically achievable. Primary standards must protect sensitive individuals, including the elderly and those already suffering from respiratory and cardiopulmonary disorders.

Secondary standards are established to protect public welfare e.g. structures, crops, animals, fabrics etc.

National Ambient Air Quality Standards (NAAQS) are for six criteria pollutants. These pollutants are carbon monoxide, lead, nitrogen dioxide, ground-level ozone, sulphur dioxide and particulate matter.

Table: National Ambient Air Quality Standards (NAAQS)

S.No.	Pollutants	Averaging Time	Federal Primary	Federal Secondary	Most Relevant Effects
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1.	Carbon monoxide (CO)	8 hr	9 ppm (10 mg/m ³)	None	Aggravation of angina pectoris; decreased Exercise tolerance; possible risk to fetuses
		1 hr	35ppm(40 mg/m ³)	None	
2.	Nitrogen dioxide (NO ₂)	Annual mean	0.053 ppm (100 micro gm/m ³)	Same	Aggravation of respiratory disease; atmospheric discoloration
		1hr	None	None	
3.	Ground level ozone (O ₃)	8 hr	0.08 ppm (155 micro gm/m ³)	Same	Decreased pulmonary function; surrogate for eye irritation; materials and vegetation damage
4.	Sulfur dioxide (S O ₂)	Annual mean	0.03 ppm (80micro gm/m ³)	None	Wheezing, shortness of breath, chest tightness; plant damage and odor
		24 hr	0.14 ppm (365micro gm/m ³)	None	
		3 hr	None	0.50ppm	
		1 hr	None	0.25ppm	
5.	PM 10	Annual	50 micro gm/m ³	Same	Exacerbation of respiratory mean disease symptoms; excess deaths; visibility
		24 hr	150 micro gm/m ³	Same	
6.	PM 2.5	Annual	15 micro gm/m ³	Same	
		24 hr	65 micro gm/m ³	Same	
7.	Lead (Pb)	1 month 3 months	None 1.5 micro gm/m ³	None Same	Impaired blood formation; infant development.

The NAAQS forms the basis for an air pollution index, called the Pollutant Standards Index (PSI). It has been developed by the EPA and used by all metropolitan areas in the United States to report to the public an overall assessment of a given day's air quality. The PSI integrates air quality data

for five of the criteria pollutants (except lead) into a single number that represents the worst daily air quality in an urban area.

Human health effects-Bhopal gas tragedy

The Bhopal gas tragedy took place in Bhopal, Madhya Pradesh, India on December 3rd, 1984. Which was due to the gas released from a pesticide manufacturing plant (Carbamate pesticide) of Union Carbide Factory, caused death of approximately 3200 persons. The methyl isocyanate was used for the manufacturing of Carbamate pesticide. MIC was stored under refrigeration in an underground tank at the Bhopal plant site. The leakage of MIC was due to the refrigeration units were out of order for several months and the two safety measures for neutralizing leaking MIC viz. the vent gas scrubber and the flare tower were also non-functional.

Exposures to MIC lead to chest tightness, breathing troubles and eyeache and also cyanide generation in the body, which ultimately turns fatal.